Blockchain Lab #1

HYOUNG JOONG KIM
Kookmin University

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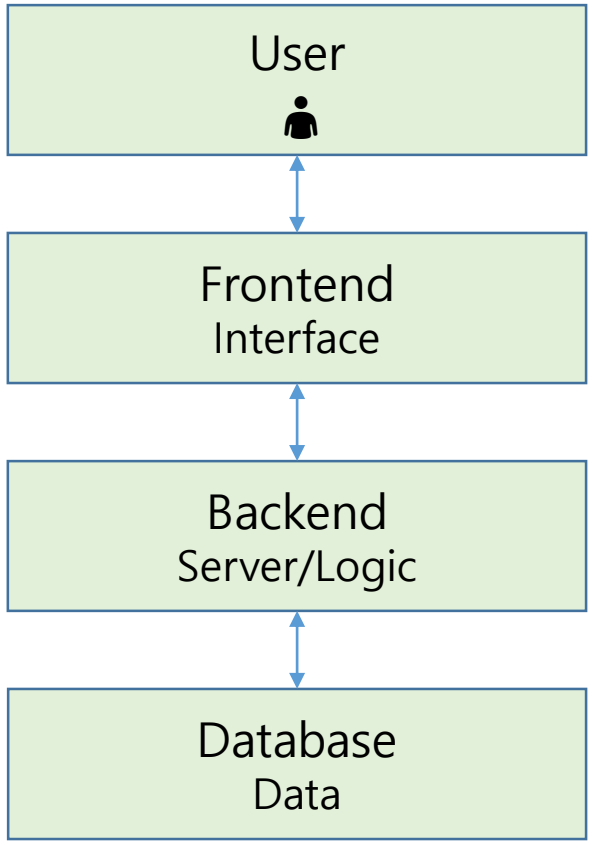
Keywords

1. Ethereum
2. Smart Contract
3. dApp
4. Vibe Coding
   - "A development approach where instead of writing code directly, you convey your intent (vibe) to AI to build software“
   - **Andrej Karpathy**, co-founder of OpenAI, first used the term in February 2025 on X (Twitter): "*You fully give in to the vibes, forget that the code even exists... I don’t need to understand how the code works*“
5. Agent

Full-Stack

- What is Full-Stack?
  - "Handling both Frontend + Backend together"



	Frontend	Backend
Role	Screen layout, UI/UX	Data processing, business logic
Visibility	Users can see it	Users cannot see it
Examples	Buttons, forms, animations	Login handling, payments, DB queries
Key Tech	React, HTML, CSS	Node.js, Python, Java

- In the past, you needed separate frontend and backend developers — but now:
  - **With AI + Next.js, a single person can build a full-stack service from scratch.**

JavaScript

- JS = JavaScript
  - JS is simply an abbreviation for JavaScript.

- What is JavaScript?
  - It was originally created **purely for making web browsers interactive and dynamic.**

HTML → Structure of a webpage (skeleton)
CSS → Design of a webpage (style)
JS → Behavior of a webpage (movement)

- **Why is JS So Popular?**
  - The **only language natively supported** by all browsers
  - One language that covers **both frontend and backend**
  - One of the languages **AI understands best** in vibe coding

Name	Meaning
JavaScript	The original language
Node.js	A runtime that lets JS run outside the browser (on servers)
Next.js	A full-stack framework built on top of React (which is JS-based)
React	A UI library built with JS
TypeScript	JS with type safety added (.ts, .tsx)

React

HTML = The restaurant building
CSS = Interior design
JS = Staff
React = Smart, well-trained staff
Next.js = Smart staff + kitchen + reservation system all-in-one

Situation	Plain JS Staff	React Staff
Updating the menu	Replaces the entire menu board	Only updates the item that changed
Managing multiple tables	Checks everything from scratch each time	Remembers and tracks each table's status
Repeating the same service	Recreates it from scratch every time	Reuses pre-built components
Avoiding mistakes	Has to manually keep track of everything	System checks automatically

Visit and see: https://upbit.com/exchange?code=CRIX.UPBIT.KRW-BTC

Lecture Schedule (First 7 Weeks)

Week	Topics	Assignment
1	수업 안내, 조교 소개, 학생 소개	MetaMask 설치 & 사용법 설명
	LLM, Vibe Coding, Git	
2	Stablecoin, Micropayment, AI Agent가 만나는 지점, Web2 vs Web3 결제 구조 비교	
	MetaMask 설치, Seed 문구 개념, EOA vs Smart Contract Account	
3	트랜잭션 구조(nonce, gas, signature), Public/Private Key 이해, Testnet, Faucet 개념	
	Sepolia ETH 받기, 지갑 간 송금, Etherscan으로 트랜잭션 추적	
4	Smart Contract란? 상태(state)와 함수(function), require, msg.sender, msg.value	
	Remix로 첫 컨트랙트 배포, “Hello Blockchain”	
5	Hardhat 프로젝트 구조, 로컬 체인 vs 테스트넷, ABI 개념	
	Hardhat + MetaMask 연결, 컨트랙트 배포 스크립트 작성	
6	Token이란? balanceOf, transfer, approve, allowance, OpenZeppelin 사용 이유	
	ERC-20 토큰 직접 배포	
7	중간고사 기간	

Git

- What is Git?
  - A system that tracks and manages every change you make to your code

Feature	Meaning
Version Control	Saves snapshots of your code at any point in time
Rollback	Made a mistake? Go back to any previous version
Branching	Work on new features without touching the main code
Collaboration	Multiple people edit the same project without conflict
History	See exactly who changed what, and when

- Without Git = Chef cooks directly on the final dish
  - One mistake → entire dish is ruined
- With Git = Chef tests on a separate plate first
  - Happy with it? Merge it into the final dish
  - Not happy? Throw it away, original is safe

- Git = The tool that tracks changes (runs on your computer)
- GitHub = The cloud platform that stores your Git repos online

Markdown

- What is Markdown?
  - Markdown is a lightweight markup language created by John Gruber in 2004.
  - It allows you to format plain text using simple symbols, which are then converted into HTML or other formatted output.
  - It's widely used in README files, documentation, blogs, and tools like GitHub, Notion, and Obsidian.

--- ## **About Me**

Hi, I'm **Alex Johnson** — a passionate full-stack developer based in San Francisco. I love creating clean, fast, and user-friendly web applications.

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- 💻 **Web Development** — React, Node.js, TypeScript
- 🎨 **UI/UX Design** — Figma, Tailwind CSS
- 📊 **Data & APIs** — REST, GraphQL, PostgreSQL
- ☁️ **Cloud & DevOps** — AWS, Docker, CI/CD

A markdown sample

What to Do Today

1. Create your Github account.
2. Create your homepage on GitHub.
3. Upload a table to GitHub showing today's Bitcoin and Ethereum prices. Fetch the coin prices from CoinGecko and populate the table with them. Document these steps as a Markdown file. Finally, ask an LLM to convert your Markdown file into an HTML file.
4. Lastly, implement a program that leverages the CoinGecko API to fetch and automatically update the Bitcoin and Ethereum prices in the table in real time.

Comparison of Tools

Feature	Remix IDE	Hardhat	Foundry
Pricing Model	Free	Free	Free
Cost Details	Completely free, open-source IDE. No subscription or usage fees.	Completely free, open-source framework. No subscription or usage fees.	Completely free, open-source toolchain.
Core Purpose	Browser-based IDE for quick prototyping, learning, and simple deployments.	A professional, flexible development framework with excellent debugging and a massive plugin ecosystem.	An extremely fast, Rust-based toolchain focused on performance, Solidity-native testing, and features like fuzzing.
Primary Language	Solidity (UI-driven)	JavaScript/TypeScript	Solidity (for tests & scripts), Rust (for toolchain)
Best For	Absolute beginners, quick contract testing, using the built-in debugger to learn EVM.	Full-stack JavaScript/TypeScript developers, projects needing extensive plugins and mainnet forking.	Experienced Solidity developers, DeFi projects, teams prioritizing test speed and security through fuzzing.

Comparison of Tools

Tool	Position	Strengths	Pricing	For Your Workflow
Claude Code	CLI Agent (#1)	Multi-agent, 1M context, parallel processing	$200/month (Pro Max)	Terminal-based collaboration with LLM
Cursor	AI-Native IDE	Full project understanding, free model choice	$20/month	Can completely replace Remix
GitHub Copilot	IDE Plugin	Easy setup, GitHub integration, affordable	$10/month	Adds AI to your existing IDE
OpenAI Codex	CLI Agent	Specialized for complex debugging	High-cost	Complementary to Claude